SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE:CSA1105

COURSE OUTCOME COVERED

CO1: *Apply object-oriented concepts and software development life cycle models to design structured and modular software systems.* *(BL2, BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:30**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I P.Malreddy (192372015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1.Suppose a software system is developed for a **Smart Hotel Management System** where guests can book rooms, cancel reservations, and make online payments.

(a) Explain the following object-oriented concepts:

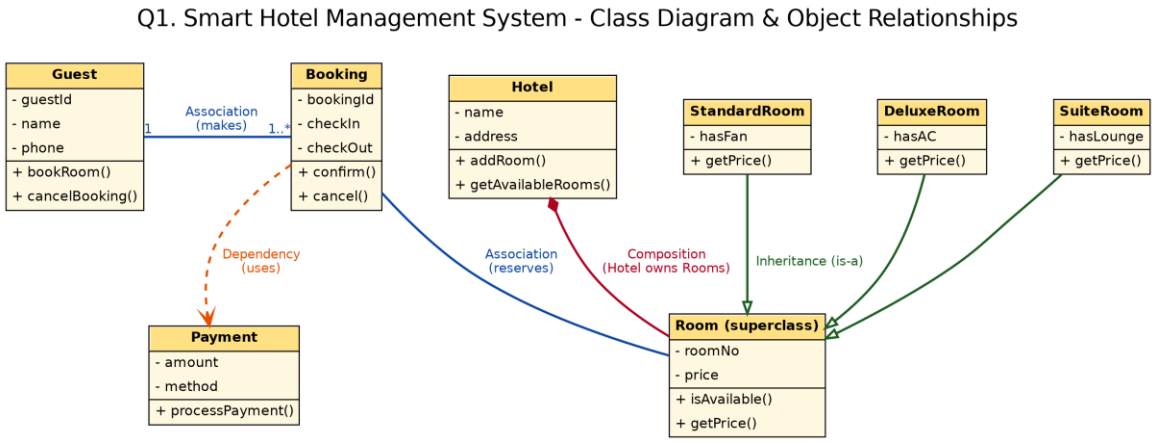
- Class
- Object
- Inheritance
- Encapsulation
- Dynamic Binding

(b) Describe the following object relationships:

- Association
- Dependency
- Composition

(c) Explain how these concepts improve software maintainability and code reusability in a hotel management system.

(5 Marks)



2.Consider developing a **Smart Education Management System**.

(a) Explain the following object-oriented design principles:

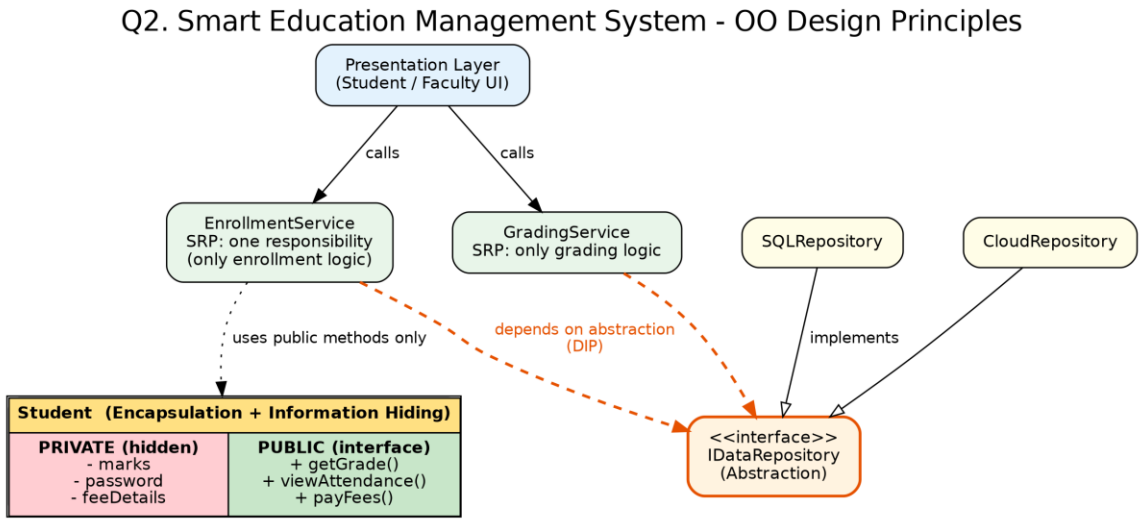
- Abstraction
- Encapsulation
- Information Hiding

(b) Describe the following SOLID principles:

- Single Responsibility Principle (SRP)
- Dependency Inversion Principle (DIP)

(c) Explain how these principles improve software scalability and maintainability.

(5 Marks)



3.Suppose a software company is developing a **Smart Pharmacy Management System**.

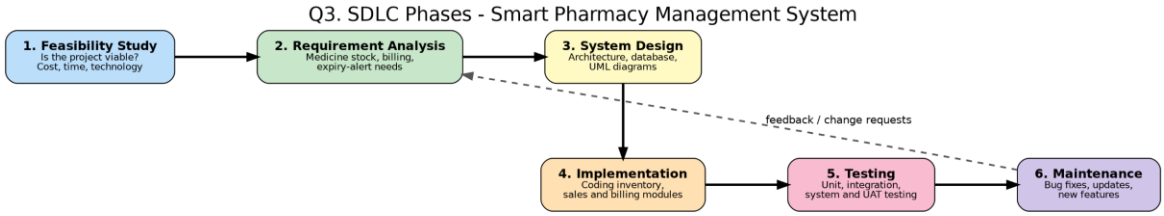
(a) Explain the phases of the Software Development Life Cycle (SDLC):

- Feasibility Study
- Requirement Analysis
- System Design
- Implementation
- Testing
- Maintenance

(b) Describe the significance of each SDLC phase.

(c) Explain how SDLC helps reduce project risks and improve software quality.

(5 Marks)



4.Consider a project for developing a **Smart E-Commerce System**, where customer requirements frequently change due to market demands.

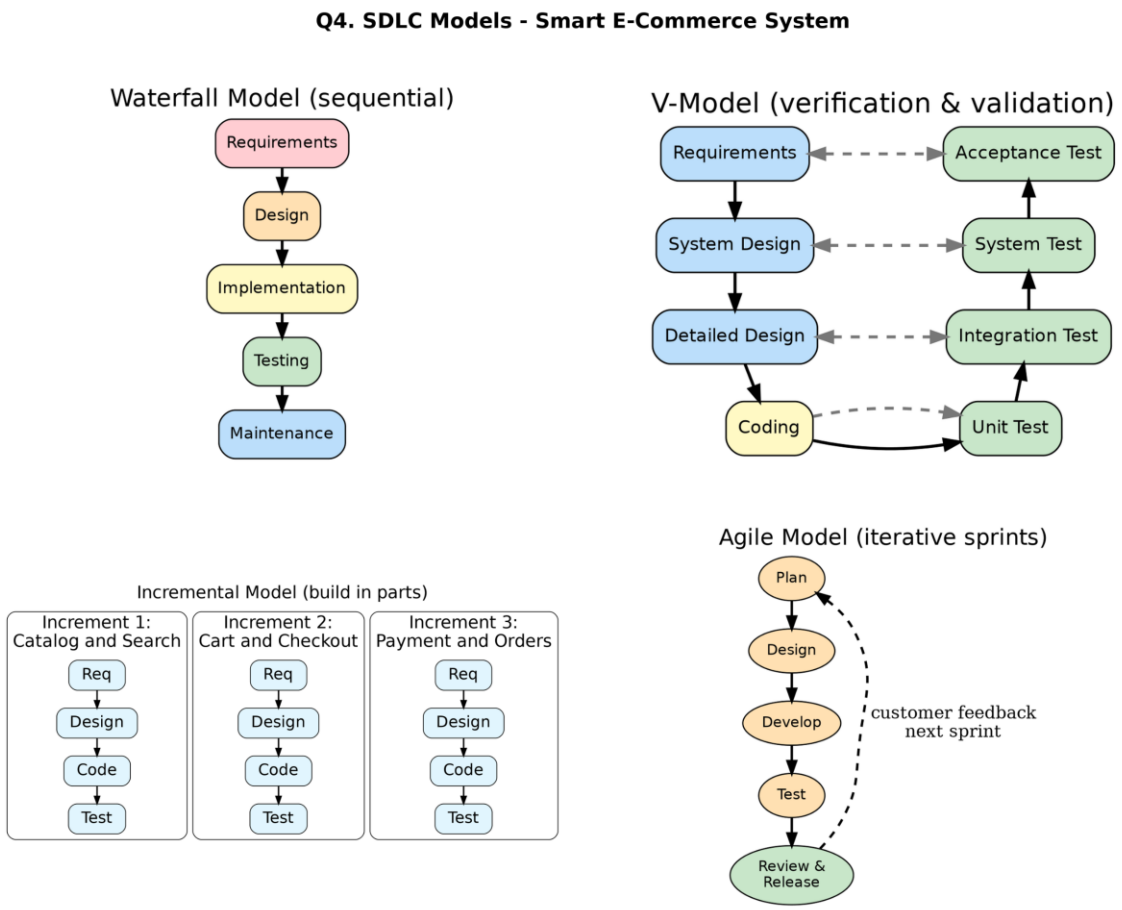
(a) Explain the following SDLC models:

- Waterfall Model
- V-Model
- Incremental Model
- Agile Model

(b) Compare the Waterfall Model and Incremental Model.

(c) Recommend the most suitable SDLC model for this project with justification.

(5 Marks)



5.Suppose a software analyst is designing a **Smart Hospital Information System**.

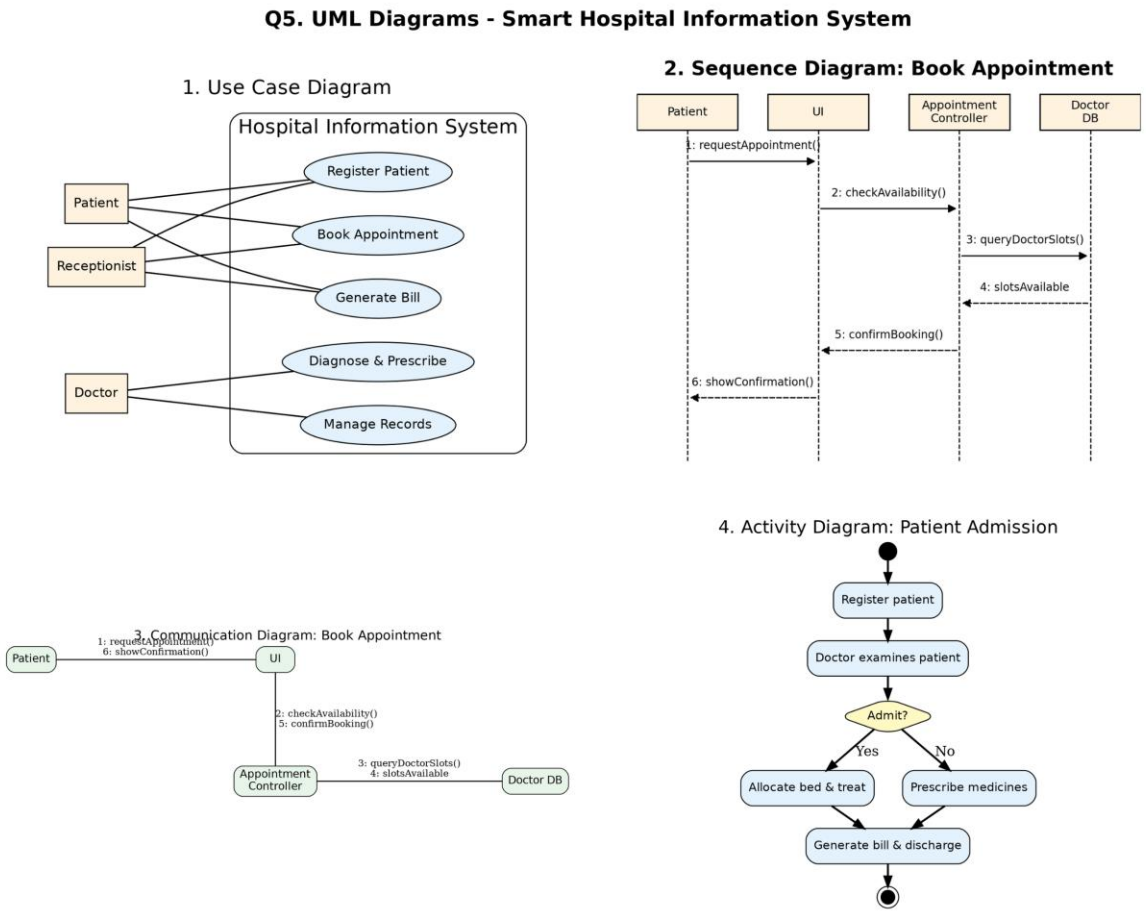
(a) Explain the role of UML in object-oriented software design.

(b) Describe the following UML diagrams:

- Use Case Diagram
- Sequence Diagram
- Communication Diagram
- Activity Diagram

(c) Explain how UML diagrams support software analysis, design, and documentation.

(5 Marks)



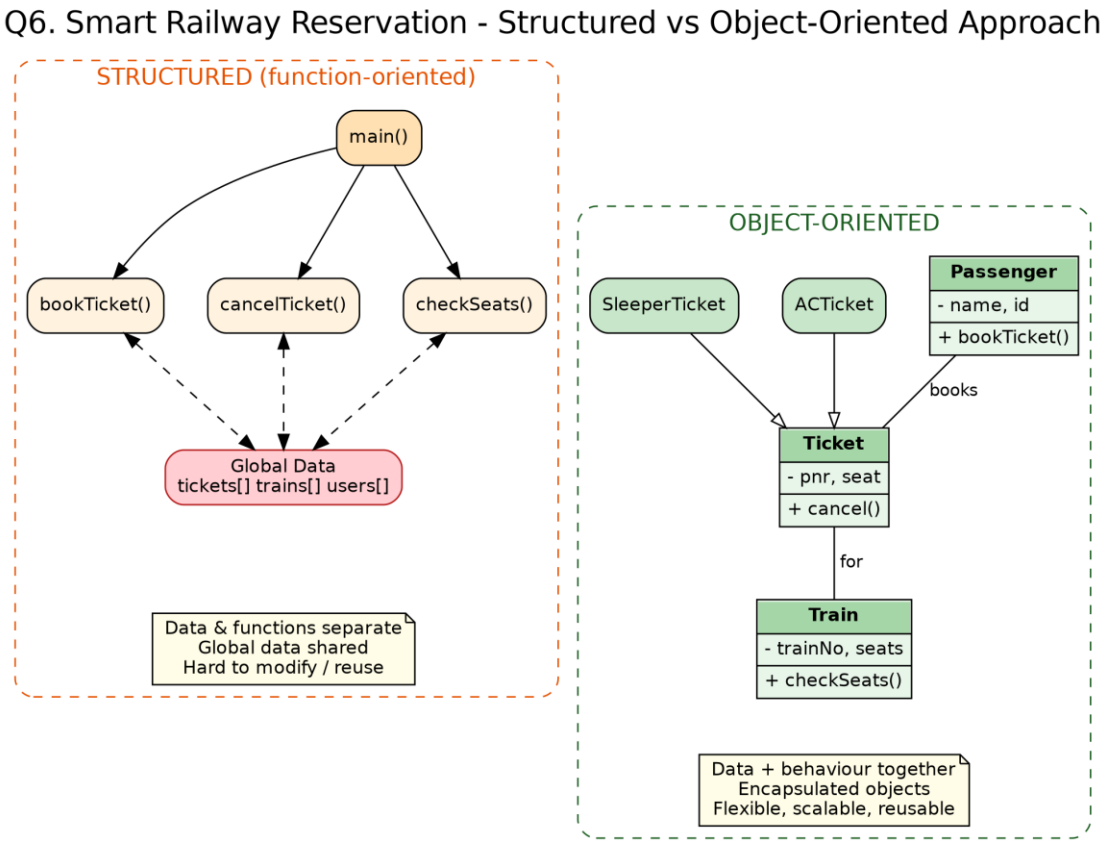
6.Consider developing a **Smart Railway Reservation System**.

(a) Explain the concept of object-oriented software design.

(b) Compare the structured software development approach with the object-oriented software development approach.

(c) Explain how object-oriented design improves software flexibility, scalability, and reusability.

(5 Marks)



RUBRICS FOR CALCULATION

SHORT ANSWER TEST

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	20%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps